INSTITUTE OF SCIENCE & TECHNOLOGY FOR ADVANCED STUDIES & RESEARCH
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Department of Computer Science (A.Y. 2025-2026)
Master of Computer Application (MCA) & Master of Science (Information Technology)
Semester - I
Subject: Practical based on Python - Beginner to Professional

Remarks:

- You might want to create a new folder with your **EnrollmentNo./YourName** for each exercise to keep them organized.

- Write a separate program to accomplish each of these exercises in juypter notebook. Save each program with a filename that follows standard Python conventions, using lowercase letters and underscores, such as **program_helloworld.py** and **simple_messages.py**

	Unit – 1
	python.org: Explore the Python home page (http://python.org/) to find topics that interest you. As you become familiar with Python, different parts of the site will be more useful to you.
1	**1-1. Hello World Typos**: Open the hello_world.py file you just created. Make a typo somewhere in the line and run the program again. Can you make a typo that generates an error??
2	**2-1. Simple Message:** Store a message in a variable, and then print that message.
Try It You r self	**2-2. Simple Messages:** Store a message in a variable, and print that message. Then change the value of your variable to a new message, and print the new message.
	2-3. Personal Message: Store a person's name in a variable, and print a message to that person. Your message should be simple, such as, "Hello Student, would you like to learn some Python today?"
	2-4. Name Cases: Store a person's name in a variable, and then print that person's name in lowercase, uppercase, and titlecase.
	2-5. Famous Quote: Find a quote from a famous person you admire. Print the quote and the name of its author. Your output should look something like the following, including the quotation marks: Albert Einstein once said, "A person who never made a mistake never tried anything new."
	2-6. Famous Quote 2: Repeat Exercise 2-5, but this time store the famous person's name in a variable called famous_person. Then compose your message and store it in a new variable called message. Print your message.
	2-7. Stripping Names: Store a person's name, and include some whitespace characters at the beginning and end of the name. Make sure you use each character combination, "\t" and "\n", at least once. Print the name once, so the whitespace around the name is displayed. Then print the name using each of the three stripping functions, lstrip(), rstrip(), and strip().

	2-8. Number Eight: Write addition, subtraction, multiplication, and division operations that each result in the number 8. Be sure to enclose your operations in print statements to see the results. You should create four lines that look like this: print(5 + 3) Your output should simply be four lines with the number 8 appearing once on each line.
Try It Your self	**2-9. Favorite Number:** Store your favorite number in a variable. Then, using that variable, create a message that reveals your favorite number. Print that message.
	2-10. Adding Comments: Choose two of the programs you've written, and add at least one comment to each. If you don't have anything specific to write because your programs are too simple at this point, just add your name and the current date at the top of each program file. Then write one sentence describing what the program does.
3	**Write a program to use Python String**
4	**Write a program to use Python List**
	4-1. Names: Store the names of a few of your friends in a list called names. Print each person's name by accessing each element in the list, one at a time.
Try It Your self	**4-2. Greetings:** Start with the list you used in Exercise 4-1, but instead of just printing each person's name, print a message to them. The text of each message should be the same, but each message should be personalized with the person's name.
	4-3. Your Own List: Think of your favorite mode of transportation, such as a motorcycle or a car, and make a list that stores several examples. Use your list to print a series of statements about these items, such as "I would like to own a Honda motorcycle."
5	**Write a program to use Python Tuple**
6	**Write a program to use Python Dictionary**
7	**Write a program to use Python Arithmetic Operators**
8	**Write a program to use Python Assignment Operators**
9	**Write a program to use Python Logical Operators**
10	**Write a python program - discount is calculated on the input amount. Rate of discount is 5%,if the amount is less than 5000,otherwise 10% .**
11	**Write a python same program with the help of if...elif....else.** **Write a python program - discount is calculated on the input amount. Rate of discount is 5%, if the amount is above than 1000 and less than 10000, and 10% if it is above 10000, else 0% discount.**
12	**Write a python program for use of Nested if.**
13	**Write a program to use Python While loop.**
14	**Write a Python program to use for Loop.**
15	**Write a Python Program to use break.**
16	**Write a Python Program to use Continue.**
17	**Write a Python Program to use Pass.**
18	**Write a python program to use of break in a for loop iterating over a list. print that list. User inputs a number, which is searched in the list. If it is found, then the loop terminates with the 'found' message**
19	**Write a program to purposefully raise Indentation Error and Correct it**
20	**Write a python program to print a number is positive/negative using if-else.**
21	**Write a Python Program to calculate the square root.**
22	**Python Program to convert temperature in Celsius to Fahrenheit.**

	fahrenheit = celsius * 1.8 + 32 celsius = (fahrenheit - 32) / 1.8
23	**Write a program in python to check a number whether it is prime or not.**
24	**Write a Python program to display all the prime numbers within an interval.**
25	**Write a python program to check whether the given number is palindrome or not.**
26	**Write a python program to find area of circle.**
27	**Write a Python program to find the largest number among the three input numbers.**
28	**Print the first 10 natural numbers using for loop range function.**
29	**Write a Python Program to print Multiplication table (from 1 to 10) in Python**
30	**Python program to print all the even numbers within the given range.**
31	**Python program to count the total number of digits in a number.**
32	**Python program that accepts a word from the user and reverses it.**
33	**Write a program to count the number of vowels present in a string.**
34	**Write a Python Program to display the Fibonacci sequence up to n-th term.**
35	**Write a program using a for loop that loops over a sequence.**
36	**Write a program using a while loop that asks the user for a number, and prints a countdown from that number to zero.**
37	**Python program to find the factorial of a given number.**
38	**Python program to check the validity of password input by users.**
39	**Python program to convert the month name to a number of days.**
40	**Write a program to count the numbers of characters in the string and store them in a dictionary data structure.**
41	**Write a program to use split and join methods in the string and trace a birthday with a dictionary data structure.**
42	**Write a program the combination of an else statement with a for statement that searches for even number in given list.**
43	**Write a program the combination of an else statement with a while statement that prints a number as long as it is less than 5, otherwise the else statement gets executed.**